



PROFILE HANDBOOK

# Programs A-I

The cartridge-free field manual for Power Glove Vision profiles.

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The **PowerGlove Vision Controller (Arduino UNO Q)** recognizes these shared gestures and sends the selected Program's gamepad mapping to RetroPie.

Bad Street Brawler supplied nine programs that configured the original Power Glove for different games. This guide explains how those programs worked and how PowerGlove Vision recreates their controls with a camera.



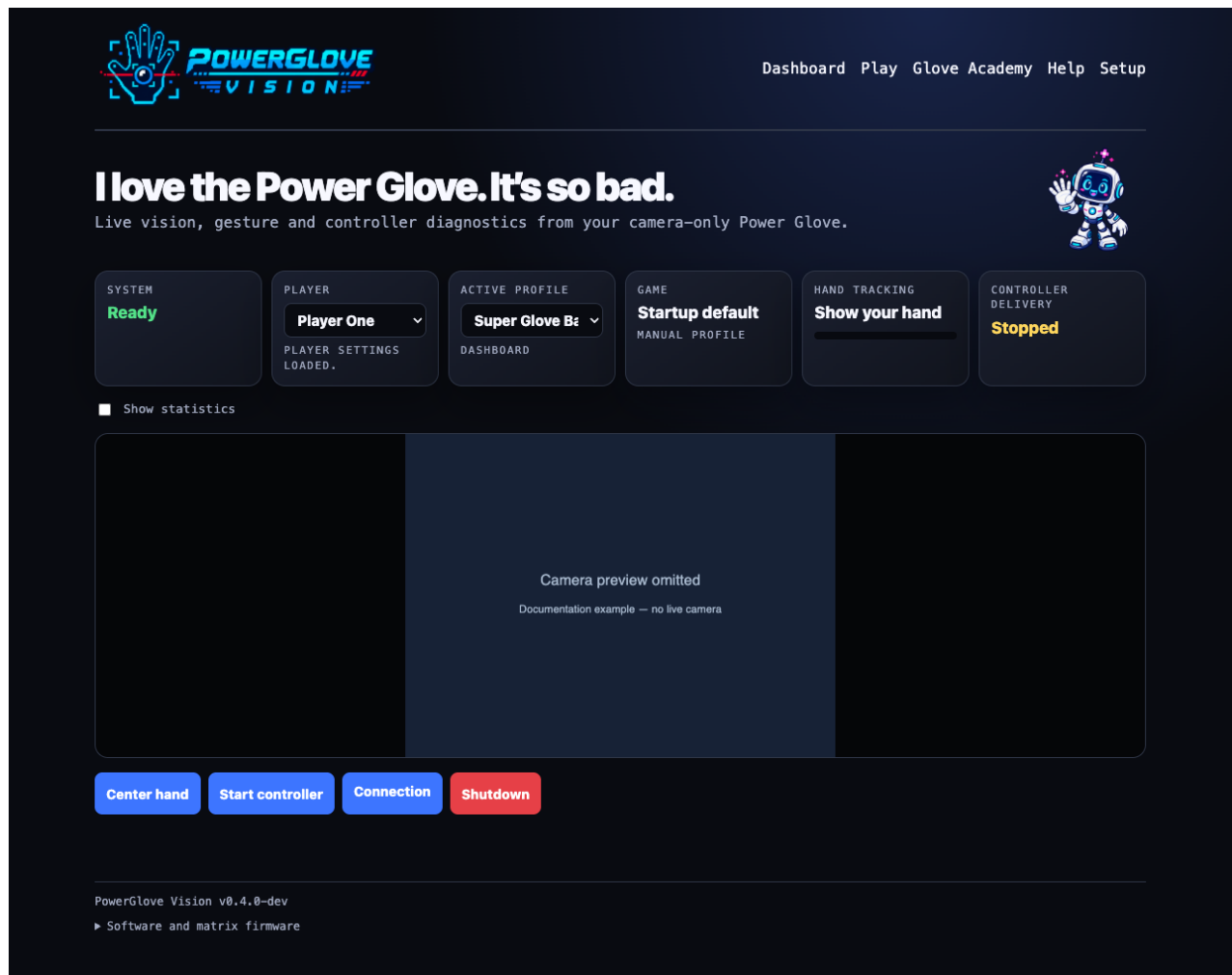
*Pixel Pal, your arcade practice buddy*

**Pixel Pal says:** Learn one gesture in Glove Academy, pick a program, then try a round. For a picture of each basic gesture, see [Gesture reference](#).

**Pixel Pal's Extra-Digit Hunt:** Some glove illustrations have five fingers plus a thumb. Count every six-digit hand once per appearance, including repeated artwork. Pixel Pal reveals the answer at the back of this guide.

## Choose a program and try it

1. Use the selector below to choose a program for your game or experiment.
2. Select that profile on Dashboard, wait for tracking, and check its gestures with controller delivery stopped.
3. Select **Start controller** when ready to play. Use the [Gameplay Guide](#) for game objectives and first-round exercises.
4. If the mapping suits another game, add its exact filename to the registry using [Register games and select profiles](#).



Dashboard profile selector and controller diagnostics

To change sensitivity without changing a program's button assignments, open **Glove Academy** -> **Tune gestures**. Pixel Pal guides the adjustment; numerical thresholds are under **Advanced**. The matrix shows **T** during tuning and **L** during the sixteen learning lessons. Each player keeps saved progress toward **Glove Master**. See the illustrated [tuning walkthrough](#).

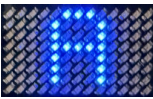
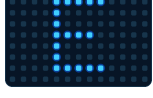
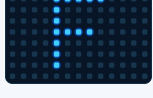
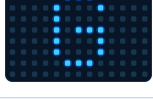
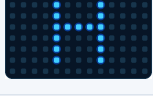
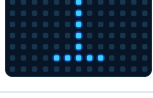
## Where the programs came from

Bad Street Brawler contained nine configuration programs labelled A through I. The player loaded one into the Power Glove, switched off the NES, swapped to a different cartridge within roughly 30 seconds, and kept using the downloaded mapping.

These programs were not replacement firmware. They were small configurations for the glove's resident gesture interpreter. Each program mapped hand position, depth, wrist angle, and finger bends to ordinary NES controller inputs. The next game therefore did not need special Power Glove support.

PowerGlove Vision keeps all nine profiles ready at once. Select one on Dashboard or let RetroPie choose it when a game launches. You do not need to open Bad Street Brawler first.

## Quick selector

| Program | See it                                                                              | Best fit                | Main controls                                                   |
|---------|-------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------|
| A       |    | Pinball                 | Two finger flippers, wrist tilt, combined-flipper mode          |
| B       |    | Joust                   | Steer by position; curl a finger to flap                        |
| C       |    | Gyruss                  | Rotate by wrist angle; fire and bomb gestures                   |
| D       |    | Challenge mode          | Reversed directions with thumb/index buttons                    |
| E       |    | Defender II             | Ship movement, fire, smart bomb, evasive movement               |
| F       |    | Sesame Street 1-2-3     | Open-hand Yes and closed-hand No                                |
| G       |  | Gun Smoke               | Walk by position; combine index curl and a forward push to fire |
| H       |  | Training / general play | Familiar controls with pulsed buttons                           |
| I       |  | Knight Rider / driving  | Wrist steering, throttle, brake, and turbo                      |

## Menu gestures shared by every program

| Gesture                                                | See it                                                                              | Controller result                  |
|--------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------|
| Hold a V sign steadily                                 |  | Start or pause after half a second |
| Briefly show a thumbs-up with the other fingers closed |  | Select                             |

These menu poses suppress A/B attacks while they form. Some programs can still produce directional output from wrist, depth, or finger gestures. Keep your hand near its calibrated resting position, and return to a relaxed open hand after the

command is recognized.

# Program cards

## A - Pinball rig

| Gesture                            | See it                                                                            | Controller result                 |
|------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------|
| Curl the index finger              |  | Right flipper / A                 |
| Curl the thumb                     |  | Left flipper / Up                 |
| Roll the wrist left or right       |  | Tilt / B                          |
| Pull the hand away from the camera |  | Toggle combined-flipper behaviour |

Use this profile for pinball tables and games that benefit from two independent finger actions.

## B - Joust rig

| Gesture                         | See it                                                                              | Controller result                                                     |
|---------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Move the hand left or right     |  | Steer left or right                                                   |
| Curl the thumb                  |  | B button; see the <a href="#">Joust play card</a> for its in-game use |
| Curl the index or middle finger |  | Pulsed flap input                                                     |

Use this profile for Joust and any game where rhythmic, repeated presses matter.

## C - Gyruess rig

| Gesture                            | See it                                                                            | Controller result                     |
|------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------|
| Roll the wrist left or right       |  | Rotate counter-clockwise or clockwise |
| Keep the index finger straight     |  | Continuous fire                       |
| Pull the hand away from the camera |  | Launch a bomb                         |

Use this profile for circular shooters and games with rotation plus rapid fire.

## D - Mirror-world rig

| Gesture                     | See it                                                                              | Controller result                |
|-----------------------------|-------------------------------------------------------------------------------------|----------------------------------|
| Move the hand left or right |    | Reversed right or left direction |
| Raise or lower the hand     |   | Reversed down or up direction    |
| Curl the thumb              |  | First action button              |
| Curl the index finger       |  | Second action button             |

Use this profile for deliberate chaos, party challenges, or accessibility experiments that need inverted direction mappings.

## E - Defender rig

| Gesture                      | See it                                                                            | Controller result         |
|------------------------------|-----------------------------------------------------------------------------------|---------------------------|
| Move the whole hand          |  | Move the ship             |
| Curl the thumb               |  | Fire                      |
| Roll the wrist left or right |  | Smart bomb                |
| Curl the ring finger         |  | Rapid horizontal movement |

Use this profile for Defender II and multi-action shooters.

## F - Yes / No rig

| Gesture                            | See it                                                                              | Controller result |
|------------------------------------|-------------------------------------------------------------------------------------|-------------------|
| Close every finger into a fist     |   | No                |
| Move an open hand in any direction |  | Yes               |

Use this profile for Sesame Street 1-2-3 and simple choice-driven games.

## G - Gun Smoke rig

| Gesture                        | See it                                                                            | Controller result                                |
|--------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------|
| Move the whole hand            |  | Walk using horizontal and vertical hand movement |
| Curl the index finger          |  | Fire                                             |
| Roll the wrist                 |  | Add left or right movement                       |
| Curl the thumb and ring finger |  | Suppress all ordinary controller output          |

A forward push sends B. Combine it with index curl (A) to send A+B.

Use this profile for Gun Smoke and shooters with movement plus directional fire.

## H - Training rig

| Gesture                   | See it                                                                              | Controller result         |
|---------------------------|-------------------------------------------------------------------------------------|---------------------------|
| Move the whole hand       |  | Conventional directions   |
| Curl the thumb            |  | Pulsed B                  |
| Curl the index finger     |  | Pulsed A                  |
| Return the hand to center |  | Release directional input |

Use this profile for learning the system or giving an unmapped game a sensible general-purpose starting point.

## I - Driving rig

| Gesture                         | See it                                                                            | Controller result |
|---------------------------------|-----------------------------------------------------------------------------------|-------------------|
| Roll the wrist left or right    |  | Steer             |
| Curl the index finger           |  | Throttle          |
| Lower the hand                  |  | Brake             |
| Push the hand toward the camera |  | Turbo             |
| Curl the thumb                  |  | Auxiliary action  |

Use this profile for Knight Rider and driving games that need steering, speed, and one extra action.

## How PowerGlove Vision selects a program

Choose the current profile on Dashboard; choose the saved startup profile on Setup. Automatic selection matches the complete ROM filename, including its extension but excluding its folder path, against [/etc/powerglove/games.json](#) on RetroPie. Matching ignores letter case.

The launch hook sends an authenticated profile request. The PowerGlove Vision Controller releases held controls, changes the mapping, reuses the saved calibration, and acknowledges the new profile on its blue matrix. If no valid calibration is saved, it collects an initial reference while you hold your open hand still in a comfortable resting position. It uses 24 geometrically valid observations; repeating the same center, distance, and wrist pose produces a similar rather than bit-for-bit identical reference.

```
JSON
{
  "games": {
    "Joust (USA).zip": "program_b",
    "Gyruss (USA).nes": "program_c",
    "Gun.Smoke (USA).7z": "program_g"
  }
}
```

Merge entries into the existing [games](#) object; do not replace other registered games. The matrix displays **A** through **I**. When an unknown game starts, the launch hook turns gesture control off so the previous game's mapping does not remain active.

## Practise safely

1. Open <http://UNO-Q-NAME.local:8088/learn>.
2. Keep your whole hand visible. Select **Center hand** on first use or when your resting position produces unwanted movement, then hold still.
3. Move slowly until the intended gesture is recognized consistently.
4. Open Dashboard to compare hand motion with generated controller output.
5. Start controller delivery only when ready to play.

Glove Academy mode works without RetroPie and automatically starts the camera while suppressing controller output. This also works while **Gestures off** is the selected profile. Leaving Glove Academy restores the selected mode, so it is a safe place to build muscle memory without changing the saved or active selection.

## What remains special

The nine profiles emit useful conventional controller inputs, so ordinary NES cores can use them today. Bad Street Brawler's Glove Zap also works through standard inputs: its dedicated profile emits a short simultaneous Left + Right pulse. FCEUmm must allow opposing directions for that game. Richer native glove integration remains separate work; PowerGlove Vision preserves analogue and finger data for it and does not modify ROM images.

For installation, secure pairing, RetroArch mapping, and troubleshooting, see the [Installation Guide](#).

## Project note

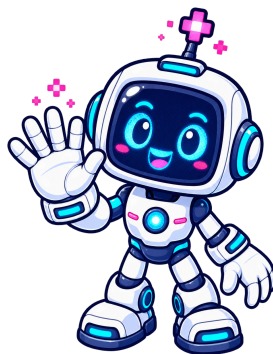
PowerGlove Vision is an independent MIT-licensed hobbyist project by Iain Bennett. Nintendo, NES, Power Glove, Bad Street Brawler, and other marks belong to their respective owners. No ROM images are distributed with this project.

### Personal sensitivity and practice

Glove Academy has sixteen lessons, including **Glove Zap**, **Pull Back**, both wrist rolls, close hand, and menu guard. In **Tune gestures**, tell Pixel Pal whether a hand is new, a gesture is difficult, a gesture happens accidentally, or movement feels off-centre. The wizard waits for one second of clear, stable tracking before the user starts its countdown. Pose, direction, and roll steps take two seconds; Glove Zap and Pull Back use three short motion-and-return repetitions so a still near or far hand cannot teach the motion. A preview must pass two uses and releases plus three neutral seconds before it can be saved.

Saved thresholds apply across profiles, including these programs. Movement controls use separate activation and release thresholds, while program-specific button pulses and mappings remain in effect. Advanced values and selective reset remain available without exposing them in the normal family workflow. Practice, personalization, preview, and diagnostics pause controller delivery.

## Pixel Pal's Extra-Digit Hunt answer



*Pixel Pal reveals the Extra-Digit Hunt answer*

**Pixel Pal's answer: 3 six-digit hands.**

They appear once each in Program B's finger-curl illustration, Program E's finger-curl illustration, and Program F's closed-hand illustration. Every appearance counts, even when the same artwork returns.